Discrete Mathematics Syllabus

CEX723 • Pamela Scott

Science and Engineering • Computer Engineering

1. Learning Objectives & Taxonomy

ID	Learning Objective	Bloom's Level
LO1	Apply propositional logic and truth tables to evaluate the validity of logical arguments in computational systems.	Applying
LO2	Construct formal mathematical proofs using direct, contrapositive, contradiction, and induction methods.	Creating
LO3	Analyze set operations, functions, and relations to model data structures and computational hierarchies.	Analyzing
LO4	Utilize counting principles and graph theory to solve problems related to network design and algorithmic complexity.	Applying
LO5	Evaluate Boolean algebra expressions and number theory applications for digital logic and cryptography.	Evaluating

Overall Course Cognitive Depth: Analyzing and Evaluating

2. Weekly Instructional Plan

Wk	Subject and Activities	Outcomes
1	**Introduction to Discrete Mathematics, course orientation, and mathematical language for computer engineering** Course Objectives / Learning Outcomes, Syllabus • Ice Breakers	—
2	**Logic, propositions, and truth tables** Face to Face Lectures • Think-Pair-Share • Quizzes	LO1
3	**Methods of proof: direct proof, contrapositive, contradiction, and counterexample** Mathematical Proofs • Challenge Problems, Assignments / Homework	LO2
4	**Sets, set operations, and Venn diagrams** Lecture Notes • Concept Maps • Problem Sets	LO3
5	**Functions, relations, and properties of relations** Peer Learning • Group Exercises • Quizzes	LO3
6	**Counting principles, permutations, and combinations** Study Questions • Collaborative Study Sessions • Assignments / Homework	LO4
7	**Recurrence relations and mathematical induction** Recitation Sessions • Tutorial Sessions • Midterm Exam	LO2
8	**Graph theory basics: graphs, paths, cycles, and connectivity** Real-World Cases • Concept Tests • Quizzes	LO4
9	**Trees, spanning trees, and rooted tree structures** Flow Charts • Think-Pair-Share • Assignments / Homework	LO4
10	**Boolean algebra and switching logic** Simulations • Peer Coding • Lab Quizzes	LO5
11	**Relations revisited: partial orders, equivalence relations, and partitions** Group Discussions • Concept Maps • Problem Sets	LO3
12	**Counting with generating ideas, pigeonhole principle, and introductory recurrence applications** Study Questions • Escape Room, Quizzes	LO4
13	**Graph traversal, shortest paths, and basic algorithms on graphs** Workshops • Active Learning Exercise • Assignments / Homework	LO4
14	**Finite automata and formal language foundations** Video Lectures • Discussion Forums • Online Exam	LO4
15	**Comprehensive review, integration of course concepts, and final exam preparation** Practice Exams • Office Hours • Assignments / Homework	—
16	**Final Exam** Final Exam	—

3. Skill Acquisitions by Category

Domain	Specific Skill	Related Wk
Hard	Formal Logical Reasoning	2, 11
	Mathematical Proof Construction	3, 7
	Algorithm Analysis	7, 14
	Network Modeling	8, 9, 10
Soft	Analytical Problem Solving	1, 3, 12
	Collaborative Technical Communication	5, 6, 11
Meta	Computational Thinking	1, 15

4. Proposed Micro-Credential Framework

Credential Name	Dimensions	Depth	Align.	Mapping
Foundations of Computational Logic	Foundational Knowledge	Proficiency	W2, 11	#FormalLogicalReasoning #BooleanAlgebra
Applied Graph Theory & Network Analysis	Application	Mastery	W8, 9, 10	#NetworkModeling #AlgorithmAnalysis

5. Assessment & Mapping

Activity Name	Week	Weight	Bloom's	Outcomes
Quizzes	2	10%	Remembering	LO1
Assignments	3	15%	Applying	LO2
Midterm Exam	7	25%	Analyzing	LO1, LO2, LO3
Lab Quizzes	10	10%	Applying	LO5
Online Exam	14	10%	Analyzing	LO4
Final Exam	16	30%	Evaluating	LO1, LO2, LO3, LO4, LO5